

Method of Moment Estimation

Definition (Moment of a random variable)

Let X be a random variable and $c \in \mathbb{R}$ a scalar. Then:

The k th moment of $X = E[X^k]$
and the k th moment of X (about c) $= E[(X - c)^k]$

Note. Moments of a random variable tells alot about its distribution for example the first moment of X : $\mu = E[X]$, and the second moment of X about μ : $\sigma^2 = \text{Var}(X) = E[(X - \mu)^2]$.

Definition

Let X be a random variable, and $c \in \mathbb{R}$ a scalar. Let $X_1, \dots, X_n$ be iid samples from X . The k th sample moment of X is

$$\text{The } k\text{th sample moment of } X = \frac{1}{n} \sum_{i=1}^n X_i^k$$

$$\text{and The } k\text{th sample moment of } X \text{ (about } c) = \frac{1}{n} \sum_{i=1}^n (X_i - c)^k$$

Note: The first sample moment is just the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, and the second sample moment about the sample mean is the sample variance

$$S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Intuitively the idea behind Method of Moments (MoM) estimation is that: to find a good estimator, we should have the true and sample moments match as best we can. That is, I should choose the parameter θ such that the first true moment $E[X]$ is equal to the first sample moment $\bar{X}$.

(Depending on how many parameters we are looking for, we will explain this in the next slides) Let's first solve an example before formalizing that.

Eg 1. Let's say $X_1, X_2, \dots, X_n$ are iid from $X \sim \text{Exp}(\theta)$. (These values may look like $x_1 = 3.21$, $x_2 = 5.11$, $x_3 = 4.33$, etc.) What is the MoM estimator of θ ?

The first moment = The first sample moment

The true mean = The sample mean

$$\begin{aligned} E[X] &= \frac{\sum_{i=1}^n x_i}{n} \\ X &\sim \text{Exp}(\theta) \\ \Rightarrow E[X] &= \frac{1}{\theta} \\ \frac{1}{\theta} &= \frac{\sum_{i=1}^n x_i}{n} \end{aligned}$$

solve for θ
 $\Rightarrow$

$$\hat{\theta} = \frac{1}{\bar{x}} = \frac{n}{\sum_{i=1}^n x_i}$$

Method of Moments Estimation

Let $X = (X_1, \dots, X_n)$ be iid samples from probability mass function $p_X(t; \theta)$ (if X is discrete), or from density $f_X(t; \theta)$ (if X continuous), where θ is a parameter (or vector of parameters). We then define the method of moments (MoM) estimator $\hat{\theta}_{MoM}$ of $\theta = (\theta_1, \dots, \theta_k)$ to be a solution (if it exists) to the k simultaneous equations where, for $j = 1, \dots, k$, we set the j th (true) moment equal to the j th sample moment:

$$E[X] = \frac{1}{n} \sum_{i=1}^n X_i$$

...

$$E[X^k] = \frac{1}{n} \sum_{i=1}^n X_i^k$$

Eg 2. Let's say $X_1, X_2, \dots, X_n$ are iid samples from $X \sim \text{Pois}(\theta)$. (These values might look like $X_1 = 13, X_2 = 5, X_3 = 4$, etc.) What is the MoM estimator of θ ?

since there is only one parameter to be estimated we only need one equation.

By MoM,

$$E[X] = \frac{\sum_{i=1}^n X_i}{n}$$

$$X \sim \text{Pois}(\theta)$$

$$\Rightarrow$$

$$E[X] = \theta$$

$$\hat{\theta}_{\text{MoM}} = \frac{\sum_{i=1}^n X_i}{n}$$

Eg 3. Let's say $X_1, X_2, \dots, X_n$ are iid samples from $X \sim N(\theta_1, \theta_2)$. (These values might look like $X_1 = -2.321$, $X_2 = 1.112$, $X_3 = -5.221$, etc.) What is the MoM estimator of the vector $\theta = (\theta_1, \theta_2)$ (θ_1 is the mean, and θ_2 is the variance)?

We are looking for two parameters, hence we need two equations.

By MoM,

$$\begin{cases} E[X] = \frac{\sum_{i=1}^n X_i}{n} & \textcircled{1} \\ E[X^2] = \frac{\sum_{i=1}^n X_i^2}{n} & \textcircled{2} \end{cases}$$

Before we go any further, first

$$\text{var}(X) = E[X^2] - (E[X])^2 \Rightarrow \text{var}(X) + (E[X])^2 = E[X^2] \quad \textcircled{3}$$

So we can rewrite our system of equations as

$$\left\{ \begin{array}{l} E[X] = \frac{\sum_{i=1}^n X_i}{n} \\ \text{var}(X) + (E[X])^2 = \frac{\sum_{i=1}^n X_i^2}{n} \end{array} \right.$$

$$\left\{ \begin{array}{l} \hat{\theta}_1 = \frac{\sum_{i=1}^n X_i}{n} = \bar{X}_n \end{array} \right.$$

$$\left\{ \begin{array}{l} \hat{\theta}_2 = \frac{\sum_{i=1}^n X_i^2}{n} - (\bar{X}_n)^2 \end{array} \right.$$

Eg 4. Let $X_1, \dots, X_n$ be a random sample from

$$f_{\theta}(x) = \begin{cases} (\theta + 1)x^{\theta} & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the method of moments estimator (MOM) for θ .

we are looking for one parameter, hence we need one equation.

$$E[X] = \frac{\sum_{i=1}^n X_i}{n} \quad (*)$$

we should first solve for $E[X]$.

$$E[X] = \int_{-\infty}^{\infty} x f_{\theta}(x) dx = \int_0^1 (\theta+1) x^{\theta+1} dx = \frac{\theta+1}{\theta+2} x^{\theta+2} \Big|_0^1$$

$$E[X] = \frac{\theta+1}{\theta+2}$$

Now we plug the value of $E[X]$ in (*)

$$\frac{\hat{\theta}+1}{\hat{\theta}+2} = \frac{\sum_{i=1}^n x_i}{n} \Rightarrow \frac{\hat{\theta}+1}{\hat{\theta}+2} = \bar{x}$$

$$\Rightarrow \hat{\theta}+1 = \bar{x} \hat{\theta} + 2\bar{x} \Rightarrow \hat{\theta}(\bar{x}-1) = 1-2\bar{x}$$

$$\hat{\theta} = \frac{1-2\bar{x}}{\bar{x}-1} \quad \text{if } \bar{x} \neq 1$$